

Minutes Document 25

Date:14/03/2024

Time: 1:30 PM - 2:30 PM

Location: Djanogly Library, Jubilee Campus, University of Nottingham

Present (in order of attendance):

- **Andreas** - Team Admin
- **Dixon** - Team Member
- **Mateusz** - Git Admin
- **Humais** - Git Helper
- **Daniel** - Deputy Git Master
- **Martin** - Team Leader

Agenda:

- **Welcome and introduction (1 minutes):**
 - Call the meeting to order
 - Converse with present team members
 - Briefly introduce the purpose of the meeting
- **What needs to be discussed (10 minutes):**
 - Making sure everyone is on track with what needs to be done for the next sprint
 - Taking a look at how the backend can use the data itself - where they can get the data from the consensus or artificial data.
- **What we need to do for Thursday (45 Minutes):**
 - **ANDREAS** - Writing up MM documentation and doing Sprint points (and all the UI tasks)
 - **MARTIN** - Ensuring communication between the company contact and Neelima, create pitch decks for these meetings
 - **MATEUSZ** - Making sure all these tasks are added to git so we can keep track of what needs to be done, in development etc etc - **GIT ADMIN** (and the tasks given to Jacob and Daniel)
 - **AARON** - Communicating with Andreas on the UI elements that must be done.
 - **JACOB & DANIEL** - Communicating with the backend team on the tables and if they are appropriate for the data being used by the MLM
 - **HUMAIS & DIXON** - Communicating with the UI team on user verification and with the MLM team on if the tables have been made correctly with the correct relationships for the MLM to use.

- UI Team:

- Refine the existing UI so that on the dashboard, it contains a form and an area where the user will be able to see their results and predictions - Andreas and Aaron are making a list of all the edits that need to be made to the webapp for it to look perfect

- Communicate with Humais for updates in the Backend - if anything needs to be changed in the fields

- For user location access, you can take a look at the HTML5 Geolocation API to get users' current locations with their consent. This will require SSL to ensure security.

- Look into implementing email notifications using SMTP protocol or third-party services like SendGrid, integrated via backend APIs, triggered upon the completion of emissions calculations or when the user signs into the app.

- Make sure this is all documented!

- Need to create a Sign-Up Page for next sprint

- MLM Team:

- Begin gathering datasets related to vehicle emissions, public transport, and geospatial data - can look at producing the artificial data using some sort of python script.

- Initial model considerations could involve looking into regression models for emissions estimation based on transport modes and distances - Look at implementing a vector database using tools like Elastic search for efficient search and matching of commute patterns.

- Make sure this is all documented!

- Backend Team:

- Try and finish up implementing the user verification with the AWS database

- Look into AWS Lambda for serverless API functions that interact with the front end and AI components - how the MLM will

- Look at creating the tables that are outlined in the report given by Martin - Design the database schema focusing on normalisation to reduce redundancy and improve data integrity. This includes creating tables for users, journeys etc with proper foreign keys for relational integrity.

- Make sure this is all documented!

Next Meeting: Thursday 21st March - Time: 1:00PM - Place: Computer Science Building

Submitted by Andreas Sergiou